Modeling of kilonova light curves from neutron star mergers and possible constraints on nuclear physics

by

Diego Fernando Hernández Porras

Submitted to the Department of Physics
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ABSTRACT

Neutron stars are compact objects whose hydrostatic equilibrium is determined by the equation of state (EOS) of the dense matter in their interiors, which makes them natural laboratories for nuclear physics at densities inaccessible to terrestrial experiments. When two neutron stars merge, part of their mass is ejected and, as demonstrated by the gravitational wave event GW170817 and its associated kilonova, this ejected material is a site of rapid neutron capture (r-process) nucleosynthesis. The yield of heavy nuclei in the ejecta depends on the properties of neutron rich nuclei far from stability, including nuclear masses, decay rates, neutron capture rates, fission properties among others much of them unknown. Since nuclear physics play a crucial rol on the volution of observables, understanding the impact of relying on theoretical nuclear physics results is essential to correctly interpret current and future multimessenger observations.

This work presents a detail analysis of the impact of different theoretical $\alpha-$ and β^-−decay rates in the kilonova evolution. The nucleosynthesis was calculated using the nuclear reaction network WinNet across an ensemble of different initial thermodynamic conditions parametrized by initial electron fraction ($Y_{e,0}$), expansion timescale (τ) and initial entropy per baryon (s_0). The kilonova light curve were calculated base on current semianalytical models based on radiative transfer simulations. The results shows that the heating rate from radioactive decays deviate from the widely accepted power law behavior depending on the β^-−decay prescription even for different initial conditions. In addition the set of nuclei responsible for such behavior was identified and the description of their decay chains are presented.

Ths next paragraph may change as results from GR simulations are been made The results on the impact of theoretical rates described above are then tested in a merger simulation where the resulting kilonova is now described as a super position of different ejecta components. The amount of ejected mass and its velocity are then extracted from a general relativistic simulation which was compute using Einstein Toolkit framework.

Thesis supervisor: Neelima Govind Kelkar
Title: Professor of Physics

Acknowledgments

Este trabajo respresenta el final de una de las etapas mas bonitas de mi vida. Llevaré con mucho cariño el recuerdo de todos mis amigos a donde me lleve la motivación.

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Chapter 1

Introduction

Binary neutron star mergers (BNSMs) are among the most energetic events in the universe, releasing tremendous amounts of energy through gravitational waves, neutrinos, and electromagnetic radiation. Following the merger of two neutron stars, a fraction of their mass is ejected through strong tidal interactions. The ejected material is expected to consist of extremely neutron rich matter, within which rapid neutron captures (r-process) lead to the formation of heavy nuclei far from stability. Consequently, these mergers are considered one of the primary sites for the synthesis of the heaviest elements in the Universe [1,2]. As the freshly synthesized nuclei decay toward stability, their radioactive energy release drives the thermodynamical evolution and produce a thermal electromagnetic transient [3]. Such transient is called Kilonova and will be the main topic of this thesis since they encode crucial information about the physics of the merger such as the abundance pattern of elements and the amount of material.

1.1 Kilonovae in a nutshell

Kilonovae are closely related to the nuclear and atomic phenomena inside the ejecta. As heavy nuclei decays towards stability part of the energy is released in form of high energy photons according to the specific transitions inside the nuclei. Also decay products such as α and β particles, which carry part of the energy of the decay as well, will interact with surrounding charged particles primarily through coulomb interaction which will produce photons that wil steal part of that energy. So effectively the relevant particles will be just photons.

At early stage the density is still high, so photons can not easily scape and interact with several charged particles before reaching the surface. As photons interact matter their energy spectrum approaches the one of a black body, this process is called thermalization and the same process can be applied to all the other decay products [4]. The emission at such stages then is well described as a black body with a temperature evolution govern by the heating rate from radioactive decays. As the density decreases bound states like atoms are being form in a process called electronic reconfiguration. After this point the thermalization of the photons depends on the emission and absorption spectrum of heavy elements. As a consequence kilonova light curves encapsule information the abundance of certain nuclei

with some specific emission and absorption lines. For example, the presence of lanthanides increases the opacity of the ejected material in an order of magnitude, so a kilonova with lanthanides will thermalize more efficiently and will shape the kilonova into a long lived thermal spectrum [5]. Since the abundance of lanthanides along with the rest of nuclei are determined by nuclear physics inputs, such as reaction rates, decay rates and masses, a detailed treatment of their uncertainties is needed for the correct interpretation of current and future observations. This is were this research work sits and will be developed later in the text.

In the folowing sections, we will first give the reader a brief historical introduction of the kilonova. Then we will present the modern models of kilonova with its limitations. And, by the end of this chapter, we will state the objective of the reseach, explain the methodology and explain the whole structure of this thesis.

1.2 A brief history of the kilonova

Until the last decade, the astrophysical origin of roughly half of the nuclei heavier than iron remained one of the major open problems of nuclear astrophysics. Big Bang nucleosynthesis accounts for the primordial abundances of the lightest nuclides (D, ^{3}He, ^{4}He and ^{7}Li), while hydrostatic and explosive burning in stars and supernovae produces the elements from carbon up to the iron peak and some heavier elements [6,7]. The issue was that beyond iron, the binding energy per nucleon decreases and Coulomb barriers grow, so heavy nuclei are built mainly by neutron captures rather than by charged particle fusion. However, the two main channels were known [8]. The first channel is the slow (s) process, neutron captures are slower than β-decays. The second is the rapid (r) process, in which captures are much faster than β-decays and drive matter far to the neutron-rich side of stability. The s-process is now well understood and operates mainly in asymptotic giant branch (AGB) stars and, for the lighter heavy elements, in massive stars [9]. The conditions required by the r-process have also been characterized.Neutron densities $n_n \gtrsim 10^{20}\,\mathrm{cm}^{-3}$, or equivalently suitable combinations of electron fraction, entropy and expansion timescale in the ejecta [10]. But, the site where these conditions occur, however, remained elusive.

For several decades the favoured candidate was the core-collapse supernova, in particular the neutrino driven wind from the newborn neutron star [11]. Modern simulations, however, find this wind neither neutron rich nor entropic enough to produce the heaviest r-process nuclei [12], so ordinary supernovae contribute at most a weak r-process, producing elements up to about Sr-Ag. Observations point in the same direction. The large star-to-star scatter of europium in metal-poor stars [13], the strong r-process enrichment of the ultra-faint dwarf galaxy Reticulum II [14], and the low abundance of live ^{244}Pu in deep-sea sediments [15] all indicate that the heavy r-process elements are made in rare events with large yields, rather than in frequent ordinary supernovae.

Neutron star mergers on the other hand matches those requirements. Early theoretical studies had proposed that during the merger of compact objects, tidal forces lead to the ejection of neutron-rich material [16]. This ejecta was expected to be very neutron-rich material and decompress rapidly. Ideal for r-process, enabling the synthesis of heavy neutron-rich nuclei. It was then proposed that the radioactive decay of freshly synthesized nuclei

would power a short-lived optical transient. In 1998 Li and Paczyński proposed a simple model (LP98) to calculate that possible electromagnetic counter part. The model predicted a light curve with a bolometric luminosity peak of approximately $10^{44}\ erg\ s^{-1}$ with a duration of ≈ 1 day after the merger depending on some free parameters like expansion velocity, ejected mass and an effective constant for the radioactive heating [17].

1.3 Modern models of Kilonova and observations

The LP98 model assumes an analytical energy generation rate of the form $\dot{\varepsilon}(t) \propto t^{-1}$ and models the electromagnetic emission as a blackbody spectrum. However, these assumptions may not fully capture the physical complexity of the event. As we discuss before the γ-rays produced by nuclear decays can escape without thermalizing, leading to deviations in the predicted emission spectrum. In addition, the actual energy generation rate may deviate from the assumed analytical form, making a more detailed treatment of the spectral evolution of the emission a challenging task.

The first attempt to construct a realistic model of this phenomenon was presented by Metzger et al. 2010 [3], who performed self-consistent calculations of the optical and near-infrared counterparts with two major improvements. First, they implemented nuclear reaction network calculations to evolve the abundances and compute the corresponding energy generation rate, $\dot{\varepsilon}(t)$, using both experimental and theoretical reaction and decay rates. Secondly, they performed one-dimensional Monte Carlo radiative transfer simulations, assuming opacities derived from bound–bound transitions of iron (Fe), with corrections applied to the ionization energies, in order to model the resulting light curves. The first results demonstrated an abundance pattern close to the solar system r-process abundances. Also, the energy generation rate was found to follow a power-law dependence, $\dot{\varepsilon}(t) \propto t^{-\alpha}$ with $\alpha \sim 1.1 - 1.3$, remarkably robust against variations in the electron fraction, ($Y_e \sim 0.05 - 0.35$), thermodynamic trajectories, and nuclear properties along the r-process path. The second key result demonstrated that the light curves evolve significantly different from that predicted by the simple blackbody approximation. The simulated light curves exhibited a characteristic "bump-like" behavior on different timescales across various wavelength bands and a peak luminosity three orders of magnitude greater than a typical nova with the latter being the reason why the term kilonova was first introduced.

However, radiative transfer simulations face major challenges due to the highly complex opacities arising from the dense line spectra of heavy r-process nuclei. This "opacity problem" has been addressed through several approaches. For instance, in [18], atomic structure models for representative elements were employed to compute radiative transition rates for tens of millions of lines, enabling more realistic estimates of the opacities for r-process material. A different approach was done in [19] where the opacity was estimated by taking into account electron scattering, free-free, bound-free and bound-bound transitions using analytical formulas. These advances led to the identification of distinct kilonova components based on the lanthanide content of the ejecta [20,21]. Material with a low electron fraction ($Y_e \leq 0.22$) is expected to produce lanthanide-rich ejecta with high opacity, resulting in a red, longer-lived transient commonly referred to as the red kilonova. In contrast, material ejected with a higher electron fraction ($Y_e \geq 0.22 - 0.3$), typically from the polar regions, evolves

into lanthanide-poor ejecta with lower opacity, producing a short-lived, blue optical transient known as the blue kilonova.

Nevertheless, all previous efforts remained purely theoretical and computational, as no observational data were available to validate the models. This situation changed on August 17, 2017, when the Advanced LIGO and Advanced Virgo gravitational-wave detectors made their first observation of a binary neutron star merger (BNSM) [22] with the subsequent identification of the electromagnetic counterparts in the same location, which supported the identification of the event as a BNSM. An approximately 100 s long gravitational-wave signal, GW170817, was followed by a short gamma ray burst (sGRB) GRB 170817A and an optical transient named, SSS17a/AT 2017gfo [23], found in the host galaxy NGC 4993. The source of the signal was detected across the electromagnetic spectrum in the X-ray, ultraviolet, optical, infrared, and radio bands—over hours, days, and weeks [24].

The electromagnetic transient SSS17a/AT 2017gfo was of great significance, as it provided the first compelling observational evidence for the nucleosynthesis of heavy nuclei through the r-process. As shown in [25], the observed emission was well reproduced by a three-component kilonova model. A lanthanide-poor blue component with an opacity of $\kappa = 0.5\ cm^2 g^{-1}$, ejecta mass $M_{ej} \approx 0.020 M_\odot$ and velocity $V_{ej} = 0.27c$; an intermediate (purple) component with $\kappa = 3\ cm^2 g^{-1}$, $M_{ej} \approx 0.047 M_\odot$ and $V_{ej} = 0.15c$; and a lanthanide-rich red component $\kappa = 10\ cm^2 g^{-1}$), $M_{ej} \approx 0.011 M_\odot$ and $V_{ej} = 0.14c$. Meanwhile, numerous efforts have been made to improve previous kilonova models, motivated by the prospect of obtaining high-resolution data from future observations with the James Webb Space Telescope (JWST). But performing fully consistent general relativistic simulations that include a realistic treatment of nuclear matter with proper neutrino transport, followed by detailed nucleosynthesis calculations in the ejecta and subsequent radiative transfer simulations to reproduce the observed light curves, remains computationally demanding. Consequently, several practical semi-analytical formulations have been proposed to approximate key physical processes and reduce the computational cost while retaining the essential physics of the problem.

One of the cornerstones of analytical modeling was presented in [26], which introduced a simplified model not intended to replace full multi-group radiative transfer simulations, but rather to serve as a toy model for qualitatively understanding the evolution of the transient. More refined analytical approaches were proposed shortly thereafter. For instance, [27] developed a model capable of reproducing multi-band light curves by assuming a time-dependent effective opacity and a uniform power-law energy generation rate. Some models also account for the impact of ejecta morphology on the observed emission, introducing more realistic prescriptions for non-spherical geometries and their influence on the angular dependence of the kilonova light curves [28].

The present panorama of kilonova modeling extends beyond purely analytical formulations and is now better represented by semi-analytical frameworks, such as the one introduced in [29]. This framework provides a practical approach for generating broadband light curves that account for multiple ejecta components and non-spherical geometries, while incorporating time- and composition-dependent heating rates, thermalization efficiencies, and opacities.

Such advances in kilonova modeling have opened new possibilities for studying distinct features in the light curves that may contain additional information about the composition of the ejecta. It has been theorized that the late time kilonova emission, occurring weeks to months after the merger, can reveal the radioactive signatures of specific heavy nuclei, rather

than the power-law heating that dominates the early time emission [30]. Isotopes such as 223Ra ($t_{1/2} = 11.43$ d), 225Ac ($t_{1/2} = 10.0$ d) with $\alpha-$decay and 254Cf with spontaneous fission, have been identified as potential dominant contributors to the late-time heating [31]. These nuclei release large amounts of energy per decay and the α particles thermalize more efficiently than typical β-decay products. Therefore, these nuclei are expected to produce detectable bumps or departures from a smooth power-law decline in the bolometric light curve. However, identifying such "fingerprints" requires sufficiently accurate heating-rate predictions from nucleosynthesis calculations and sensitive, multi-epoch, wide-band observations.

In March 2023, the electromagnetic transient AT2023vfi, associated with the gamma-ray burst GRB 230307A, was identified as the second confirmed observation of a kilonovae [32]. This signal has provided an early opportunity to apply these ideas observationally [33]. JWST follow-up analyses indicate that AT2023vfi's early evolution resembles AT2017gfo during the first week, and detailed spectral/photometric modeling has been used to place constraints on the presence and abundances of superheavy nuclei in the ejecta. The work presented in [34] reports upper limits and tentative abundance estimates for isotopes such as 222Rn, 223Ra, 224Ra and 225Ac. These results illustrate both the promise and the current limitations: while JWST data can probe late-time radioactive contributors, robust identification of specific isotopes requires careful propagation of nuclear-physics (decay-rate and fission-yield) and opacity uncertainties. In this context, one of the goals of this work is to assess whether a more accurate treatment of decay-rate uncertainties can improve the constraints on the abundance of heavy nuclei.

1.4 Objectives and thesis structure

As mention before, the main objective of this work is to evaluate the sensibility of kilonova light curves to choice of theoretical decay rates prescription. We constructed an ensemble of initial thermodynamics trajectories possibles within the ejecta. For each initial condition we ran three nuclear reaction network calculations with three different β^--decay prescription. Then with the results for the abundance and the heating rate we modeled the kilonova. We found consistent deviations form the power-law behavior across the ensemble of initial conditions. We also identify the nuclei responsible for such behavior along with their decay chains.

The thesis is divided in 3 main parts. The introductory part are chapters 1 and 2. The next chapter will develop the physics needed to understand neutrons stars and binaries neutrons stars. Chapters 3, 4 and 5 Will present a basic the theoretical framework regarding the three stages of the merger simualtion, the general realtivistic hydrodynamic part, the nucelar reaction network evolution and the construction of the kilonova light curve from emission models. The last stage, chapters 6, 7 and 9, will explain the details of the construction of the initial thermodynamic trajectories, the setup of the nuclear reaction network calculations with their results and the construction of the kilonova light curves.

Chapter 2

From Neutron stars to Kilonova

Kilonovae is going to be our main topic of study through the whole thesis. Every chapter and section was build with the objective to develop the intuition on all the processes that takes part in the kilonova. We will first present the neutron star with its dense matter in the general relativity description. The we will move to the problem of finding two neutron star in a close orbit and present a little discussion on why the existing of such configurations are not trivial. Then we will describe the different phases of the merger with the different mechanism of emission, gravitational waves, short gamma ray bursts and our favorite the kilonova. We will conclude with a detailed and current understanding of this last one.

2.1 The Neutron star

Neutron stars contain the densest matter known to exist in a stable state. In their outer layers, nuclei coexist with a degenerate electron gas. With increasing density, electron captures make the nuclei progressively more neutron-rich until, at the neutron-drip density $n_{\rm drip} \simeq 2.5 \times 10^{-4}$ fm^{-3} ($\rho \simeq 4 \times 10^{11}$ g cm^{-3}), neutrons are no longer bound to the nuclei and form a free neutron gas [35]. At about half the saturation density the difference between inside and outside of a nucleus vanish effectively becoming a continuum of nucleons which marks the transition from crust to core. Their core can reach several times the nuclear saturation density $n_{\rm sat} \simeq 0.15$ fm^{-3} [36]. At the same time the β equilibrium favors neutrons instead of protons, this leads to a neutron dominated composition.

Their hydrostatic equilibrium is determined by the equation of state (EOS) through the Tolman-Oppenheimer-Volkoff equations plus rotational corrections. A free, degenerate neutron gas can support stars of at most $\simeq 0.7\, M_\odot$ [37]. But, since the matter inside the star is expected to be dominated by neutrons really close to each other, the EOS will have to account the interactions between neutrons. At such densities neutrons interact strongly and experiments a repulsive short-range interaction, which in relativistic mean-field models is mediated by ω-meson exchange. Through this link between microphysics and astrophysical observables, neutron stars have become natural laboratories for strongly interacting matter at densities that cannot be reached in terrestrial experiments.

However the microscopic theory of this matter, quantum chromodynamics (QCD), cannot at present provide a complete description of the EOS in the relevant density range. This is

because at the baryon chemical potentials of neutron-star cores, QCD is non-perturbative, and lattice QCD is hampered by the fermion sign problem at finite baryon density. Controlled calculations exist only in two limits, chiral effective field theory describes nuclear matter up to about 1-2 $n_{\rm sat}$ [38], and perturbative QCD becomes reliable only at densities of order 40 $n_{\rm sat}$, far above those realized in neutron stars [39]. In the intermediate regime, where neutron-star cores reside, the EOS is described by phenomenological models whose predictions must be confronted with astrophysical observations.

As a consequence, a description from first principles of neutron star cores remains unknown. Matter at several times $n_{\rm sat}$ may consist of nucleons, may contain hyperons, or may undergo a transition to deconfined quark matter, either through a smooth crossover or through a first order hadron-quark phase transition [40].

2.2 Formation of binary neutron stars that merge

For two neutron stars to merge, three conditions must be met. Both stars must form, which requires two core-collapse supernovae so the binary must survive both explosions. Then the two remnants must end up close enough for gravitational-wave (GW) emission to bring them into contact in less than a Hubble time. For typical neutron-star masses, the initial separation must be smaller than about 5 $R_\odot$. This is two to three orders of magnitude smaller than the radii that the progenitor stars reach as supergiants. The central problem of binary-neutron-star formation is therefore how to bring the compact remnants of two massive stars to separations far smaller than the stars themselves once were, without unbinding the system on the way.

In the standard isolated channel, a double neutron star (DNS) descends from a binary of two OB stars that are massive enough to end their lives in a core-collapse SN, see figure 2.1. If the initial binary are separated enough the descending DNS will not be close enough to merge. If, on the contrary, the initial star are close, they will interact via stable or unstable mass transfer. Once one of the stars reach the end of its live the star will expands transferring part of its mass to their companion. If after the supernova the two stars remain bounded they will become a high-mass X-ray binary (HMXB). When the second star expands it will bolt the first star in a phase called common envelope [41]. In that stage the dynamical friction of the motion of the first star inside the second one will cause the necessary loss of angular momentum to bring closer the cores. This phase is not stable so eventually the envelope will fall to the stars ofr get ejected leaving behind the first NS and the He core of the second one that eventually will die in what is called a ultra-strippend SN [42]. finally the system will results in a DNS close enough to merge. The sequence of events is reviewed in [43].

2.3 Gravitational-wave-driven inspiral and the maximum separation

Once both neutron stars have formed, the only process that changes the orbit appreciably is the emission of gravitational radiation. Here I will show an estimate of the time to the

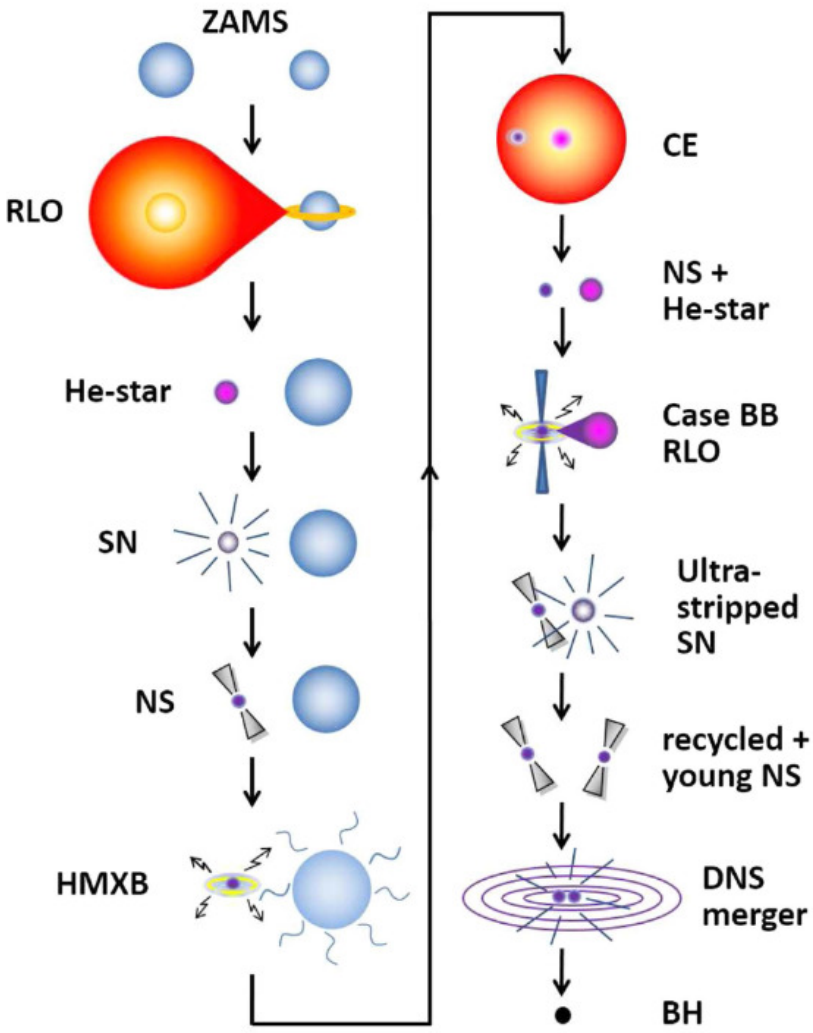


Figure 2.1: Double neutron star (DNS) formation from two OB stars as presented in [43]. ZAMS refers to zero-age main sequence, RLO roche lobe overflow, SN supernova, HMXB high mass X ray binary and CE common envelope

merger at leading (quadrupole) order and the maximum initial separation that allows a merger within the age of the Universe.

Consider two point masses m_1 and m_2 on a circular orbit of separation a and angular frequency ω, with total mass $M = m_1 + m_2$ and reduced mass $\mu = m_1 m_2/M$. Kepler's third law gives $\omega^2 = GM/a^3$. In the centre of mass frame, the relative coordinate is $\mathbf{x}(t) = a\,(\cos\omega t, \sin\omega t, 0)$, and the trace-free mass quadrupole is $Q_{ij} = \mu\,(x_i x_j - \frac{1}{3}\delta_{ij}a^2)$. Its time-dependent components oscillate at 2ω:

$$Q_{xx} = -Q_{yy} = \frac{\mu a^2}{2}\cos 2\omega t + \text{const}, \qquad Q_{xy} = \frac{\mu a^2}{2}\sin 2\omega t. \tag{2.1}$$

The quadrupole formula [44],

$$L_{\text{GW}} = \frac{G}{5c^5}\left\langle \dddot{Q}_{ij}\,\dddot{Q}_{ij}\right\rangle, \tag{2.2}$$

with $\dddot{Q}_{xx}^2 + \dddot{Q}_{yy}^2 + 2\dddot{Q}_{xy}^2 = 32\,\mu^2 a^4 \omega^6$, gives the GW luminosity

$$L_{\rm GW} = \frac{32}{5}\frac{G}{c^5}\mu^2 a^4 \omega^6 = \frac{32}{5}\frac{G^4}{c^5}\frac{(m_1 m_2)^2 M}{a^5}. \tag{2.3}$$

The energy radiated is drawn from the orbital energy $E = -Gm_1 m_2/(2a)$. Setting $\mathrm{d}E/\mathrm{d}t = -L_{\rm GW}$ gives

$$\frac{\mathrm{d}a}{\mathrm{d}t} = -\frac{\beta}{a^3}, \qquad \beta \equiv \frac{64}{5}\frac{G^3 m_1 m_2 M}{c^5}. \tag{2.4}$$

Equation (2.4) integrates to $a^4(t) = a_0^4 - 4\beta t$, so a binary with initial separation a_0 coalesces ($a \to 0$) after

$$t_{\rm c}(a_0) = \frac{a_0^4}{4\beta} = \frac{5}{256}\frac{c^5 a_0^4}{G^3 m_1 m_2 M}. \tag{2.5}$$

The steep a_0^4 dependence is the key point. Halving the separation shortens the inspiral by a factor of sixteen.

Now lets estimate the maximum separation for BNS to merger within the age of the universe. Notice that since the stars require a preliminary formation and evolution we will therefore over estimate the maximum distances between stars. Lets require $t_{\rm c} \leq t_{\rm H}$, with $t_{\rm H} \simeq 13.8$ Gyr the age of the Universe

$$a_{\rm max} = (4\beta\, t_{\rm H})^{1/4} = \left(\frac{256}{5}\frac{G^3 m_1 m_2 M}{c^5}\, t_{\rm H}\right)^{1/4}. \tag{2.6}$$

For $m_1 = m_2 = 1.35\,M_\odot$, the configuration simulated in this work, $m_1 m_2 M = 4.92\,M_\odot^3$ and $\beta = 6.08 \times 10^{27}$ cm^4 s^{-1}. Hence

$$a_{\rm max} \simeq 3.2 \times 10^6 \text{ km} \left(\frac{t}{13.8 \text{ Gyr}}\right)^{1/4} \left(\frac{m_1 m_2 M}{4.92\,M_\odot^3}\right)^{1/4} \simeq 4.6\,R_\odot, \tag{2.7}$$

about 0.02 AU. By Kepler's third law, this corresponds to a maximum orbital period $P_{\rm max} \simeq 17$ h. Equivalently, in terms of the orbital period and the chirp mass $\mathcal{M} = (m_1 m_2)^{3/5}/M^{1/5}$,

$$t_{\rm c} = \frac{5}{256}\left(\frac{G\mathcal{M}}{c^3}\right)^{-5/3}\left(\frac{P}{2\pi}\right)^{8/3} \simeq 13.8 \text{ Gyr}\left(\frac{P}{16.8 \text{ h}}\right)^{8/3}\left(\frac{\mathcal{M}}{1.175\,M_\odot}\right)^{-5/3}. \tag{2.8}$$

Two consistency checks are worth noting. At $a_{\rm max}$, the orbital velocity is $v/c = (GM/a_{\rm max}c^2)^{1/2} \approx 10^{-3}$, so post-Newtonian corrections of relative order $(v/c)^2 \sim 10^{-6}$ are negligible over almost the entire inspiral. They become important only in the last orbits. At the initial separation of the simulation in this work (45 km), the post-Newtonian expansion parameter is $x = (GM\omega/c^3)^{2/3} \approx 0.09$, and the final orbits and the merger require a full numerical solution of the Einstein equations (Chapter **??**). Tidal effects are likewise negligible until the separation is a few stellar radii.

Chapter 3

Numerical Relativistic Hydrodynamics

3.1 3+1 decomposition and evolution formulation

3.2 General-relativistic hydrodynamics in conservative form

3.3 The Einstein Toolkit: initial data, mesh refinement, atmosphere

3.4 Gravitational-wave extraction

3.5 Ejecta diagnostics and unbound-matter criteria

Chapter 4

r-Process Nucleosynthesis

4.1 Nuclear statistical equilibrium and freeze-out

4.2 The r-process: neutron-to-seed ratio, fission cycling

4.3 Reaction-network equations and numerical solution

4.4 Nuclear inputs: masses, decay rates, neutron capture, fission

4.5 Radioactive heating and thermalization

Chapter 5

Kilonova Light Curves

5.1 Li-Paczyński model

5.2 Radiative transfer and the diffusion regime

5.3 Opacities of r-process matter

5.4 Light-curve modeling

Chapter 6

Methodology on the impact of theoretical decay rates across thermodynamics trajectories

6.1 Binary neutron-star simulation

6.2 From simulation to trajectories: ejecta extraction and expansion

6.3 Initial termodinamic trajectories treatment

6.4 Nuclear reaction network setup and theoretical reaction reates

6.5 Modified Li-Paczyński model and radiative transfer simulations

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

Chapter 7

Results

7.1 Merger dynamics

7.2 Dynamical-ejecta properties

7.3 Nucleosynthesis and heating rates

7.4 Kilonova light curves

7.5 Relevant nuclei and uncertainties

7.6 Detectability with current and future telescopes

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

Chapter 8

Discussion

8.1 Comparison with previous studies

8.2 Limitations and model assumptions

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

Chapter 9

Conclusions and Outlook

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

Appendix A

Units and conventions

Appendix B

Convergence tests

Appendix C

Network and equation-of-state configuration details

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